

EE5203 L02 Study Sheet

Values, Decisions, Loops, and Arrays - Basri Erdogan

Essential question: How does one C loop turn eight raw sensor values into a validated average, a count, and a status decision — and how do you prove each intermediate value in the debugger?

What you should be able to do

- Trace assignment and arithmetic statements in execution order.
- Predict the result of integer division and remainder.
- Distinguish = (store) from == (compare).
- Write and trace **if**, **else if**, **else** decision chains.
- Trace **while** and **for** loops, including first and last counter values.
- Index an array safely and explain why `samples[8]` is invalid for `[8]`.
- Use the accumulator pattern to compute a sum and average.

Core statements

Statement	Question it answers
<code>x = 5;</code>	Nothing — it stores 5 in <code>x</code>
<code>x == 5</code>	Does <code>x</code> currently contain 5?
<code>if (c) { ... }</code>	Run the block once if <code>c</code> is true
<code>while (c) { ... }</code>	Repeat while <code>c</code> stays true (test first)
<code>for (init; test; update)</code>	Repeat with a counter over a known range
<code>while (1) { ... }</code>	Firmware main loop — repeat forever

Operators

Group	Operators	Watch out
Arithmetic	<code>+</code> <code>-</code> <code>*</code> <code>/</code> <code>%</code>	<code>10 / 3</code> is 3 (integer); <code>10 % 3</code> is 1
Shorthand	<code>x++</code> , <code>x += 5</code>	Same as <code>x = x + 1</code> , <code>x = x + 5</code>
Comparison	<code>==</code> <code>!=</code> <code><</code> <code><=</code> <code>></code> <code>>=</code>	Comparing changes neither operand
Logical	<code>&&</code> <code> </code> <code>!</code>	<code>&&</code> both true; <code> </code> at least one true

The for loop, traced

```
for (int i = 0; i < 4; i++) { /* body */ }
```

Repetition	i before body	i < 4?
1	0	yes
4	3	yes
stop	4	no

START creates `i = 0`, **TEST** runs before every repetition, **UPDATE** runs after every body. The body runs four times — `i` never reaches 4 inside the body.

Arrays

```
#include <stdint.h>

uint16_t samples[8] = { 3000, 4095, 4200, 2500,    // the lecture's data set;
                       3500, 1000, 3800, 4000 }; // the lab uses its own
```

- `uint16_t`: unsigned, 16-bit, fixed-width integer — no negative values.
- Valid indices are 0 through 7. `samples[8]` is **outside** the array.
- Loop condition is `i < 8`, never `i <= 8`.

The accumulator pattern

```
int sum = 0; // must start at zero

for (int i = 0; i < 8; i++) {
    if (samples[i] > 4095) { // clamp invalid input
        samples[i] = 4095;
    }
    sum += samples[i]; // accumulate
    if (samples[i] > 2500) {
        high_count++; // count while visiting
    }
}

int average = sum / 8; // integer division
```

One loop can clamp, sum, and count in a single pass. Every result variable needs a correct starting value **before** the loop.

Worked example (lecture data set). Trace it yourself before reading the right column; the lab set is different, so its prediction table is still yours to fill.

Result	Value	Why
<code>samples[2]</code> after the loop	4095	4200 > 4095 was true once
<code>invalid_count</code>	1	counted inside that same if
<code>high_count</code>	6	2500 > 2500 is false — > is strict
<code>sum</code>	25990	running: 3000 · 7095 · 11190 · 13690 · 17190 · 18190 · 21990 · 25990
<code>average</code>	3248	25990 / 8; the remainder 6 is discarded
<code>status_code</code>	1	3248 > 3000 → HIGH

Decision chains

```
if (average < 1500) status_code = -1; // LOW
else if (average > 3000) status_code = 1; // HIGH
else status_code = 0; // OK
```

Conditions are tested top to bottom; exactly one branch runs; testing stops at the first true condition.

Common mistakes

- Writing `=` where `==` is meant — the condition becomes an assignment.
- Forgetting to initialize an accumulator (`sum` starts with garbage).
- `i <= 8` on an eight-element array — reads/writes outside the array.
- Forgetting the update in a `while` loop — the loop never ends.
- Expecting `25990 / 8` to have a fraction — integer division gives 3248 and discards the remainder 6.
- Trusting a clean build as proof the computed values are correct.

Mastery self-check

- ☐ I can trace `x = x + 3; y *= 5; x++;` and give both final values.
- ☐ I can explain why `if (status = 1)` is a bug, not a comparison.
- ☐ I can state how many times `for (int i = 0; i < 8; i++)` runs its body.
- ☐ I can list the valid indices of `uint16_t samples[8]`.
- ☐ I can explain why `sum` must start at zero.
- ☐ I can predict `sum / 8` when `sum` is 25990.
- ☐ I can place a breakpoint inside a loop and read `i` and `samples[i]`.

Five review questions

1. What are the final values of `a` and `b` after `int a = 6, b = 4; a = a - b; b++; a *= 2;?`

2. `int x = 7;` — what is the value of the expression `x % 3`, and what is `x` afterwards?
3. A `while` loop's body never executes. What must be true of its condition, and when was it tested?
4. In one pass over an array, which of these need a starting value before the loop: the loop counter, an accumulator, a count of high samples?
5. The lab program clamps 4200 to 4095 **before** adding it to `sum`. What would the average be wrong by if it clamped after summing?

See the L02 worksheet for extended practice. Worked solutions are released after the practice window.